

Multimodal Pedagogical Friction Detector

Fusing Facial Affect · Talk Moves · LLM Reasoning
to Surface Teachable Moments at Scale

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Why this problem?

Manual classroom coding takes years of expert effort and cannot scale to real-time teacher feedback [1].

The challenge

- When students are confused, the teacher's **next move** is critical for deep learning [2]
- Existing tools are **single-modal**: transcript *or* affect — never fused with semantic reasoning
- Edu-ConvoKit & TutorCoPilot ignore student affect; affect systems ignore teacher response quality

Our goal

- Define **Pedagogical Friction**: elevated student confusion **and** a low-quality teacher move in the same 30-second window
- Fuse **vision** (DeepFace) + **language** (ELECTRA + claude-sonnet-4-6)
- Scale to **53 TIMSS lessons, 7 countries** without human labelling

Definition of Pedagogical Friction

A 30-second window where student confusion is *elevated* **AND** the teacher's talk move is *low-quality* (answer-giving, closed loops, transmission-only instruction).

Video data

- TIMSS AU1 lesson (10 min)
- 1200 frames → DeepFace @ 2 Hz
- 87 bins | 1 human positive

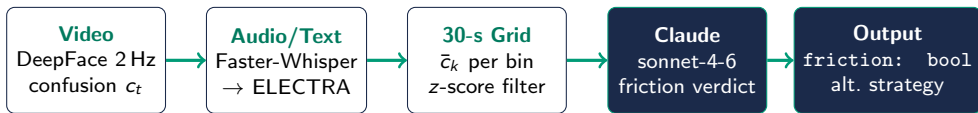
Transcript corpus

- 53 TIMSS lessons
- 28 math + 25 science
- 7 countries | 5113 bins

Annotation

- Human: 1 positive (bin 11)
- Heuristic: 31.5% flagged
- ELECTRA: 19.3% flagged

Shared 30-second temporal grid aligns vision & language streams for every lesson.



Confusion proxy (Eq. 1)

$$c_t = \sum_{e \in \mathcal{E}} w_e \cdot p_t(e)$$

$w_{\text{fear}}=0.90$, $w_{\text{sad}}=0.85$, $w_{\text{disgust}}=0.50 \dots$

Candidate filter (Eq. 2)

$$\bar{c}_k > \mu_C + z \sigma_C \wedge q_k \neq \text{high}$$

$z=1.0$ flags the top $\sim 16\%$ of bins.

Talk Move fallback: (1) ELECTRA → (2) Edu-ConvoKit → (3) Keyword heuristic

Track 1: AU1 Full Video

- 1 200 frames $\rightarrow$ DeepFace + Whisper + Claude
- 7 candidates $\rightarrow$ Claude rejects **all** (social disruption)
- Wide-angle: $\bar{c}_k=0$ every bin — **vision fails**

Track 2: 54-Lesson Scale-Up

- SLURM GPU array (NVIDIA A40)
- 5 318 bins | 1 797 flagged
- Claude confirms **21** moments (0.4%)
- Wide-angle failure confirmed

Track 3: Human Eval (87 bins)

- 1 annotator $\rightarrow$ **1 positive** (bin 11)
- Heuristic: $F1 = 0.10$
- ELECTRA: $F1 = 0.00$
- Heuristic + Claude: **$F1 = 1.00$**

Key finding

$r = -0.35$ correlation between heuristic & ELECTRA rates across 53 lessons $\Rightarrow$ **annotation methodology is a first-class design variable.**

Condition	Bins	Flag	%	F1
<i>Human evaluation (AU1, 87 bins)</i>				
Heuristic	87	20	23%	0.10
ELECTRA	87	33	38%	0.00
+ Claude	87	1	1%	1.00
<i>Transcript-only (53 lessons)</i>				
Heuristic	5113	1611	31.5%	—
ELECTRA	5113	988	19.3%	—
+ Claude	5113	211	4.1%	—

- ★ **F1 = 1.00**
Claude cuts false positives 20×; zero missed detections
- ★ **7.6× reduction**
1 611 flags → 211 confirmed friction moments
- ★ $r = -0.35$
Heuristic vs ELECTRA anti-correlated across 7 countries
- ★ **Vision limit**
Wide-angle $\Rightarrow$ DeepFace unusable; language is the primary signal

References I



Washington, DC: National Center for Education Statistics, 2003.

